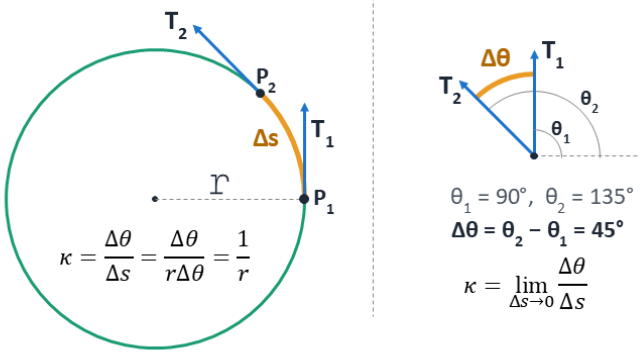
Curvature

Curvature in Angle Perspective(Scalar)

Curvature: Rate of change of the tangent angle (θ) with respect to arc length



Consider two circles with r = 1 and r = 100. The circle with r = 1 has the larger curvature. A driver moving at a constant speed along the circle with r = 100 feels a gentle turn, whereas a driver on the circle with r = 1 feels a very sharp one.

Curvature in Vector Perspective

The curvature is the rate of change unit tangent with respect to arc length. Consider a parametric curve in 2D:

$$\boldsymbol{r}(s) = \begin{bmatrix} x(s) \\ y(s) \end{bmatrix} \rightarrow \frac{d\boldsymbol{r}}{ds} = \begin{bmatrix} \frac{dx}{ds} \\ \frac{dy}{ds} \end{bmatrix}$$

This is the unit tangent vector:

$$\boldsymbol{T} = \frac{d\boldsymbol{r}}{ds}$$

The curvature is as follows:

$$\kappa = \left\| \frac{d\boldsymbol{T}}{ds} \right\|$$

Are the two perspectives same? Yes.

$$\boldsymbol{T} = \begin{bmatrix} \cos\theta \\ \sin\theta \end{bmatrix}$$

$\boldsymbol{T}$ is the unit tangent. Differentiate with respect to s:

$$\frac{d\boldsymbol{T}}{ds} = \begin{bmatrix} -\sin\theta \\ \cos\theta \end{bmatrix} \frac{d\theta}{ds}$$

$$\kappa = \left\| \frac{d\boldsymbol{T}}{ds} \right\| = \left\| \begin{bmatrix} -\sin\theta \\ \cos\theta \end{bmatrix} \frac{d\theta}{ds} \right\| = \left| \frac{d\theta}{ds} \right|$$

The magnitude of a vector $\begin{bmatrix} -\sin\theta \\ \cos\theta \end{bmatrix}$ is 1.

Curvature Formula

Curvature is not limited to circles. It is defined for any smooth curve, which we can describe parametrically. For example, a circle of radius r can be parametrized by t as $x(t) = r\cos t$ and $y(t) = r\sin t$ where $0 \leq t \leq 2\pi$. Let us now derive the curvature formula for a general parametric curve $(x(t), y(t))$.

$$\kappa = \frac{d\theta}{ds} = \frac{d\theta/dt}{ds/dt}$$

Find denominator: ds/dt

$$\frac{ds}{dt} = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}$$

$$\frac{dx}{dt} = x' \qquad \frac{dy}{dt} = y'$$

$$\frac{ds}{dt} = \sqrt{x'^2 + y'^2}$$

Find numerator: $d\theta/dt$

$$\tan\theta = \frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{\sin\theta}{\cos\theta}$$

Use $\tan\theta = \frac{dy}{dx}$

$$\frac{d}{dt}\tan\theta = \frac{d}{dt}\frac{dy/dt}{dx/dt} = \frac{d}{dt}\frac{y'}{x'} = \frac{y''x' - y'x''}{x'^2}$$

Use $\tan\theta = \sin\theta/\cos\theta$:

$$\begin{aligned}\frac{d}{dt}\tan\theta &= \frac{d}{dt}\frac{\sin\theta}{\cos\theta}\frac{d\theta}{dt} \\ &= \frac{\cos\theta\cdot\cos\theta - \sin\theta\cdot(-\sin\theta)}{\cos^2\theta}\frac{d\theta}{dt} \\ &= \frac{1}{\cos^2\theta}\frac{d\theta}{dt} \\ &= \sec^2\theta\frac{d\theta}{dt}\end{aligned}$$

Equate two results of $\frac{d}{dt}\tan\theta$:

$$\sec^2\theta\frac{d\theta}{dt} = \frac{y''x' - y'x''}{x'^2}$$

$$\begin{aligned}\frac{d\theta}{dt} &= \frac{1}{\sec^2\theta}\frac{y''x' - y'x''}{x'^2} \\ &= \frac{1}{1+\tan^2\theta}\frac{y''x' - y'x''}{x'^2} \\ &= \frac{1}{1+(y'/x')^2}\frac{y''x' - y'x''}{x'^2} \\ &= \frac{x'^2}{x'^2+y'^2}\frac{y''x' - y'x''}{x'^2} \\ &= \frac{y''x' - y'x''}{x'^2+y'^2}\end{aligned}$$

Put $\frac{d\theta}{dt} = \frac{y''x'-y'x''}{x'^2}$ and $\frac{ds}{dt} = \sqrt{x'^2+y'^2}$ into curvature formula, $\kappa = \frac{d\theta/dt}{ds/dt}$

$$\begin{aligned}\kappa = \frac{d\theta/dt}{ds/dt} &= \frac{y''x' - y'x''}{x'^2+y'^2}\cdot\frac{1}{(x'^2+y'^2)^{\frac{1}{2}}} \\ &= \frac{y''x' - y'x''}{(x'^2+y'^2)^{\frac{3}{2}}}\end{aligned}$$

Curvature in Polar Coordinate:

$$\kappa = \frac{y''x' - y'x''}{(x'^2+y'^2)^{\frac{3}{2}}}$$

$$x = r_\theta\cos\theta$$
$$y = r_\theta\sin\theta$$

$$x' = \frac{dx}{d\theta} = \frac{dr_\theta}{d\theta}\cos\theta + r_\theta\frac{d}{d\theta}\cos\theta = r'_\theta\cos\theta - r_\theta\sin\theta$$
$$y' = \frac{dy}{d\theta} = \frac{dr_\theta}{d\theta}\sin\theta + r_\theta\frac{d}{d\theta}\sin\theta = r'_\theta\sin\theta + r_\theta\cos\theta$$

$$\begin{aligned}x'' &= \frac{d}{d\theta}(r'_\theta\cos\theta - r_\theta\sin\theta) \\ &= \frac{d}{d\theta}r'_\theta\cos\theta - \frac{d}{d\theta}r_\theta\sin\theta \\ &= r''_\theta\cos\theta - r'_\theta\sin\theta - r'_\theta\sin\theta - r_\theta\cos\theta \\ &= r''_\theta\cos\theta - 2r'_\theta\sin\theta - r_\theta\cos\theta\end{aligned}$$

$$\begin{aligned}y'' &= \frac{d}{d\theta}(r'_\theta\sin\theta + r_\theta\cos\theta) \\ &= \frac{d}{d\theta}r'_\theta\sin\theta + \frac{d}{d\theta}r_\theta\cos\theta \\ &= r''_\theta\sin\theta + r'_\theta\cos\theta + r'_\theta\cos\theta - r_\theta\sin\theta \\ &= r''_\theta\sin\theta + 2r'_\theta\cos\theta - r_\theta\sin\theta\end{aligned}$$

Numerator:

$$\begin{aligned}y''x' - y'x'' = &(r''_\theta\sin\theta + 2r'_\theta\cos\theta - r_\theta\sin\theta)(r'_\theta\cos\theta - r_\theta\sin\theta) \\ &-(r'_\theta\sin\theta + r_\theta\cos\theta)(r''_\theta\cos\theta - 2r'_\theta\sin\theta - r_\theta\cos\theta)\end{aligned}$$

Step A. expand the first parenthesis

$(r'' \sin\theta + 2r' \cos\theta - r\sin\theta) \cdot (r' \cos\theta - r\sin\theta)$

1. $r'' : r''r' \sin\theta\cos\theta - r''r\sin^2\theta$
2. $r' : 2r'^2\cos^2\theta - 2r'r\sin\theta\cos\theta$
3. $r : -rr'\sin\theta\cos\theta + r^2\sin^2\theta$

Step B. expand second parenthesis:

$(r' \sin\theta + r\cos\theta) \cdot (r'' \cos\theta - 2r'\sin\theta - r\cos\theta)$
1. $r'' : r'r''\sin\theta\cos\theta + rr''\cos^2\theta$
2. $r' : -2(r')^2\sin^2\theta - 2r'r\sin\theta\cos\theta$
3. $r : -rr'\sin\theta\cos\theta - r^2\cos^2\theta$

Step A - Step B:

1. $r''r'$: $r''r'\sin\theta\cos\theta - r''r'\sin\theta\cos\theta = 0$
2. $r''r$: $(-r''r\sin^2\theta) - (r''r\cos^2\theta) = -rr''(\sin^2\theta + \cos^2\theta) = -rr''$
3. r'^2: $(2r'^2\cos^2\theta) - (-2r'^2\sin^2\theta) = 2r'^2(\cos^2\theta + \sin^2\theta) = 2r'^2$
4. $r'r$: $(-2r'r\sin\theta\cos\theta - rr'\sin\theta\cos\theta) - (-2r'r\sin\theta\cos\theta - rr'\sin\theta\cos\theta) = 0$
5. r^2: $(r^2\sin^2\theta) - (-r^2\cos^2\theta) = r^2(\sin^2\theta + \cos^2\theta) = r^2$

$\therefore y''x' - y'x'' = r^2 + 2r'^2 - rr''$

Denominator: $(x'^2 + y'^2)^{\frac{3}{2}}$

$$\begin{aligned} x'^2 &= (r'\cos\theta - r\sin\theta)^2 \\ &= r'^2\cos^2\theta - 2rr'\sin\theta\cos\theta + r^2\sin^2\theta \\ y'^2 &= (r'\sin\theta + r\cos\theta)^2 \\ &= r'^2\sin^2\theta + 2rr'\sin\theta\cos\theta + r^2\cos^2\theta \end{aligned}$$

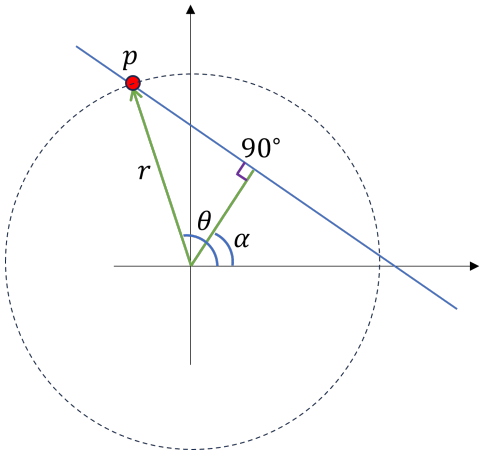
$$\begin{aligned} x'^2 + y'^2 &= r'^2\cos^2\theta - 2rr'\sin\theta\cos\theta + r^2\sin^2\theta + r'^2\sin^2\theta + 2rr'\sin\theta\cos\theta + r^2\cos^2\theta \\ &= r'^2\cos^2\theta + r'^2\sin^2\theta - 2rr'\sin\theta\cos\theta + 2rr'\sin\theta\cos\theta + r^2\sin^2\theta + r^2\cos^2\theta \\ &= r'^2 + r^2 \end{aligned}$$

Therefore, the curvature formula in polar coordinate is as follows:

$$\begin{aligned} \kappa &= \frac{|y''x' - y'x''|}{(x'^2 + y'^2)^{\frac{3}{2}}} \\ &= \frac{\left|r^2 + 2r'^2 - rr''\right|}{(r'^2 + r^2)^{\frac{3}{2}}} \end{aligned}$$

Examples

1. Prove the curvature of line is zero:



A line in polar form is most naturally described by its closest point to the origin. Drop a perpendicular from the origin to the line. Call that shortest distance d (also written p), and call the angle that perpendicular makes with the x-axis(α).

Any point P on the line can be described as (r, θ). The foot of the perpendicular is at distance d from the origin, at angle α. The angle between the direction to P and the direction of that perpendicular is $(\theta - \alpha)$. Basic trigonometry in the right triangle gives:

$$r\cos(\theta - \alpha) = d$$

$$r_\theta = \frac{d}{\cos(\theta - \alpha)} = d\sec\varphi \text{ where } \varphi = \theta - \alpha$$

from curvature in polar coordinate:

$$\kappa = \frac{\left|r^2 + 2r'^2 - rr''\right|}{(r'^2 + r^2)^{\frac{3}{2}}}$$

$$\begin{aligned}
r'_\theta &= d\frac{d}{d\theta}\frac{1}{\cos(\theta - \alpha)} \cdot \frac{d}{d\theta}(\theta - \alpha) \\
&= d\frac{0 \cdot \cos(\theta - \alpha) - 1 \cdot (-\sin(\theta - \alpha))}{\cos^2(\theta - \alpha)} \cdot 1 \\
&= d\frac{\sin(\theta - \alpha)}{\cos^2(\theta - \alpha)} \\
&= d\sec(\theta - \alpha)\tan(\theta - \alpha) \\
&= d\sec\varphi\tan\varphi
\end{aligned}$$

$$\begin{aligned}
r''_\theta &= d\sec\varphi\tan\varphi\tan\varphi + d\sec\varphi\sec\varphi\cos\varphi \\
&= d(\sec\varphi\tan^2\varphi + \sec^3\varphi)
\end{aligned}$$

Numerator: $r^2 + 2r'^2 - rr''$

$$
\begin{aligned}
r^2 &: d^2\sec^2\varphi \\
2r'^2 &: 2(d\sec\varphi\tan\varphi)^2 = 2d^2\sec^2\varphi\tan^2\varphi \\
rr'' &: d\sec\varphi\cdot d(\sec\varphi\tan^2\varphi+\sec^3\varphi) = d^2\sec^2\varphi\tan^2\varphi + d^2\sec^4\varphi \\
r^2+2r'^2-rr'' &= d^2\sec^2\varphi + 2d^2\sec^2\varphi\tan^2\varphi - d^2\sec^2\varphi\tan^2\varphi - d^2\sec^4\varphi \\
&= d^2\sec^2\varphi + d^2\sec^2\varphi\tan^2\varphi - d^2\sec^4\varphi \\
&= d^2\sec^2\varphi(1+\tan^2\varphi) - d^2\sec^4\varphi \\
&= d^2\sec^2\varphi\sec^2\varphi - d^2\sec^4\varphi \\
&= 0
\end{aligned}
$$

2. Curvature of $y = f(x)$:

$$x' = 1 \qquad x'' = 0 \qquad y' = \frac{dy}{dx} \qquad y'' = \frac{d^2y}{dx^2}$$

$$
\begin{aligned}
\kappa &= \frac{y''x' - y'x''}{(x'^2+y'^2)^{\frac{3}{2}}} \\
&= \frac{y''}{(1+y'^2)^{\frac{3}{2}}} \\
&= \frac{\frac{d^2y}{dx^2}}{\left[1+\left(\frac{dy}{dx}\right)^2\right]^{\frac{3}{2}}}
\end{aligned}
$$

Curvature in vector perspective:

$$\kappa = \frac{d\theta}{ds} = \frac{d\theta}{dx}\frac{dx}{ds}$$

Find dx/ds:

$$\boldsymbol{r} = \begin{bmatrix} x \\ f(x) \end{bmatrix}$$

where x is not arc length.

$$\frac{d\boldsymbol{r}}{dx} = \begin{bmatrix} 1 \\ f'(x) \end{bmatrix}$$

$$\left|\frac{d\boldsymbol{r}}{dx}\right| = \sqrt{1+[f'(x)]^2}$$

$$ds^2 = dx^2 + dy^2 \to ds = \sqrt{dx^2+dy^2} \to \frac{ds}{dx} = \sqrt{1+\left(\frac{dy}{dx}\right)^2}$$

$$\frac{dx}{ds} = \frac{1}{\sqrt{1+\left(\frac{dy}{dx}\right)^2}} = \frac{1}{\sqrt{1+[f'(x)]^2}}$$

Find $d\theta/ds$:

$$\begin{aligned}\boldsymbol{T} = \frac{d\boldsymbol{r}}{ds} \leftarrow \frac{d\boldsymbol{r}}{ds} &= \frac{dr}{dx}\frac{dx}{ds} \\ &= \begin{bmatrix} 1 \\ f'(x) \end{bmatrix} \frac{1}{\sqrt{1+[f'(x)]^2}}\end{aligned}$$

If the tangent makes an angle θ with the x-axis, then

$$\boldsymbol{T} = \begin{bmatrix} \cos\theta \\ \sin\theta \end{bmatrix} = \begin{bmatrix} 1 \\ f'(x) \end{bmatrix} \frac{1}{\sqrt{1+[f'(x)]^2}}$$

Therefore,

$$\cos\theta = \frac{1}{\sqrt{1+[f'(x)]^2}}$$

$$\sin\theta = \frac{f'}{\sqrt{1+[f'(x)]^2}}$$

$$\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{f'}{\sqrt{1+[f'(x)]^2}}\sqrt{1+[f'(x)]^2} = f'$$

So,

$$\theta = \tan^{-1}(f'(x))$$

Now calculate curvature

$$\kappa = \frac{d\theta}{ds} = \frac{d\theta}{dx}\frac{dx}{ds}$$

$$\frac{d\theta}{dx} = \frac{d}{dx}\tan^{-1}(f'(x)) = \frac{f''(x)}{1+f'(x)^2}$$

$$\frac{dx}{ds} = \frac{1}{\sqrt{1+[f'(x)]^2}}$$

Therefore,

$$\begin{aligned}\kappa = \frac{d\theta}{ds} &= \frac{d\theta}{dx}\frac{dx}{ds} \\ &= \frac{f''(x)}{1+f'(x)^2}\frac{1}{\sqrt{1+[f'(x)]^2}} \\ &= \frac{f''(x)}{[1+f'(x)^2]^{3/2}}\end{aligned}$$

3. Curvature of a circle

The curvature of a circle in polar coordinate($r = a, r' = 0, r'' = 0$):

$$\kappa = \frac{\left|r^2 + 2r'^2 - rr''\right|}{(r'^2 + r^2)^{\frac{3}{2}}} = \frac{r^2}{(r^2)^{\frac{3}{2}}} = \frac{r^2}{r^3} = \frac{1}{r} = \frac{1}{a}$$